CONTINUITY AND DIFFERENTIABILITY

1.	If $\frac{d}{dx}(f(x)) = \log x$, then f(x) equals : (a) $-\frac{1}{x} + C$ (b) $x(\log x - 1) + C$ (c) $x(\log x + x) + C$ (d) $\frac{1}{x} + C$
2.	The value of k for which $f(x) = \begin{cases} 3x+5, & x \geq 2 \\ kx^2, & x < 2 \end{cases}$ is a continuous function, is : (a) $-\frac{11}{4}$ (b) $\frac{4}{11}$ (c) 11 (d) $\frac{11}{4}$
3.	If $y = (x + \sqrt{x^2 - 1})^2$, then show that $(x^2 - 1)\left(\frac{dy}{dx}\right)^2 = 4y^2$.
4.	The function $f(x) = x\,\lvert x \rvert$ is (A) continuous and differentiable at $x = 0$. (B) continuous but not differentiable at $x = 0$. (C) differentiable but not continuous at $x = 0$. (D) neither differentiable nor continuous at $x = 0$.
5.	If $\tan\left(\frac{x+y}{x-y}\right) = k$, then $\frac{dy}{dx}$ is equal to (A) $\frac{-y}{x}$ (B) $\frac{y}{x}$ (C) $\sec^2\left(\frac{y}{x}\right)$ (D) $-\sec^2\left(\frac{y}{x}\right)$
6.	If $y = \sqrt{ax+b}$, prove that $y\left(\frac{d^2y}{dx^2}\right) + \left(\frac{dy}{dx}\right)^2 = 0$.
7.	If $f(x) = \begin{cases} ax+b & ; \; 0 < x \leq 1 \\ 2x^2 - x & ; \; 1 < x < 2 \end{cases}$ is a differentiable function in (0, 2), then find the values of a and b.

8.	The value of k for which function $f(x)=\begin{cases} x^2, & x \geq 0 \\ kx, & x < 0 \end{cases}$ is differentiable at $x = 0$ is : (a) 1 (b) 2 (c) any real number (d) 0
9.	If $y = \frac{\cos x - \sin x}{\cos x + \sin x}$, then $\frac{dy}{dx}$ is : (a) $-\sec^2\left(\frac{\pi}{4} - x\right)$ (b) $\sec^2\left(\frac{\pi}{4} - x\right)$ (c) $\log\left\|\sec\left(\frac{\pi}{4} - x\right)\right\|$ (d) $-\log\left\|\sec\left(\frac{\pi}{4} - x\right)\right\|$
10.	If $y = x^{\frac{1}{x}}$, then find $\frac{dy}{dx}$ at $x = 1$.
11.	If $x = a \sin 2t$, $y = a(\cos 2t + \log \tan t)$, then find $\frac{dy}{dx}$.
12.	The function $f(x) = \|x\|$ is (a) continuous and differentiable everywhere. (b) continuous and differentiable nowhere. (c) continuous everywhere, but differentiable everywhere except at $x = 0$. (d) continuous everywhere, but differentiable nowhere.
13.	If $y = \sin^2 (x^3)$, then $\frac{dy}{dx}$ is equal to : (a) $2 \sin x^3 \cos x^3$ (b) $3x^3 \sin x^3 \cos x^3$ (c) $6x^2 \sin x^3 \cos x^3$ (d) $2x^2 \sin^2 (x^3)$
14.	If $(x^2 + y^2)^2 = xy$, then find $\frac{dy}{dx}$.

15.	The function $f(x) = [x]$, where $[x]$ denotes the greatest integer less than or equal to x, is continuous at (a) $x = 1$ (b) $x = 1.5$ (c) $x = -2$ (d) $x = 4$
16.	The derivative of x^{2x} w.r.t. x is (a) x^{2x-1} (b) $2x^{2x} \log x$ (c) $2x^{2x}(1 + \log x)$ (d) $2x^{2x}(1 - \log x)$
17.	If $f(x) = \begin{cases} x^2, & \text{if } x \geq 1 \\ x, & \text{if } x < 1 \end{cases}$, then show that f is not differentiable at $x = 1$.
18.	Find the value(s) of 'λ', if the function $f(x) = \begin{cases} \dfrac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.
19.	Differentiate $\sec^{-1}\left(\dfrac{1}{\sqrt{1-x^2}}\right)$ w.r.t. $\sin^{-1}(2x\sqrt{1-x^2})$.
20.	If $y = \tan x + \sec x$, then prove that $\dfrac{d^2y}{dx^2} = \dfrac{\cos x}{(1-\sin x)^2}$.
21.	The number of discontinuities of the function f given by $f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ e^x, & \text{if } 0 \leq x \leq 1 \\ 2-x, & \text{if } x > 1 \end{cases}$ is : (A) 0 (B) 1 (C) 2 (D) 3

22.	Let $y = f\left(\frac{1}{x}\right)$ and $f'(x) = x^3$. What is the value of $\frac{dy}{dx}$ at $x = \frac{1}{2}$? (A) $-\frac{1}{64}$ (B) $-\frac{1}{32}$ (C) -32 (D) -64
23.	If $y = \log\sqrt{\sec\sqrt{x}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi^2}{16}$ is : (A) $\frac{1}{\pi}$ (B) π (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
24.	If $x = 3\cos\theta$ and $y = 5\sin\theta$, then $\frac{dy}{dx}$ is equal to : (A) $-\frac{3}{5}\tan\theta$ (B) $-\frac{5}{3}\cot\theta$ (C) $-\frac{5}{3}\tan\theta$ (D) $-\frac{3}{5}\cot\theta$
25.	The greatest integer function defined by $f(x) = [x]$, $1 < x < 3$ is not differentiable at $x =$ (A) 0 (B) 1 (C) 2 (D) $\frac{3}{2}$
26.	If $y = (\sin^{-1} x)^2$, then find $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx}$.
27.	If $y^x = x^y$, then find $\frac{dy}{dx}$.

28.	If $x \sin(a+y) - \sin y = 0$, prove that $\dfrac{dy}{dx} = \dfrac{\sin^2(a+y)}{\sin a}$
29.	Find $\dfrac{dy}{dx}$, if $y = (\cos x)^x + \cos^{-1}\sqrt{x}$.
30.	If $(\cos x)^y = (\cos y)^x$, then $\dfrac{dy}{dx}$ is equal to : (a) $\dfrac{y \tan x + \log(\cos y)}{x \tan y - \log(\cos x)}$ (b) $\dfrac{x \tan y + \log(\cos x)}{y \tan x + \log(\cos y)}$ (c) $\dfrac{y \tan x - \log(\cos y)}{x \tan y - \log(\cos x)}$ (d) $\dfrac{y \tan x + \log(\cos y)}{x \tan y + \log(\cos x)}$
31.	If $x = a \cos\theta + b \sin\theta$, $y = a \sin\theta - b \cos\theta$, then which one of the following is true ? (a) $y^2\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + y = 0$ (b) $y^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$ (c) $y^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - y = 0$ (d) $y^2\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - y = 0$

32.	Function f is defined as $$f(x) = \begin{cases} 2x+2, & \text{if } x < 2 \\ k, & \text{if } x = 2 \\ 3x, & \text{if } x > 2 \end{cases}$$ Find the value of k for which the function f is continuous at x = 2.
33.	Which of the following statements is true for the function $f(x) = \begin{cases} x^2+3, & x \neq 0 \\ 1, & x = 0 \end{cases}$? (A) f(x) is continuous and differentiable $\forall\, x \in \mathbb{R}$ (B) f(x) is continuous $\forall\, x \in \mathbb{R}$ (C) f(x) is continuous and differentiable $\forall\, x \in \mathbb{R} - \{0\}$ (D) f(x) is discontinuous at infinitely many points
34.	Let f(x) be a continuous function on [a, b] and differentiable on (a, b). Then, this function f(x) is strictly increasing in (a, b) if (A) $f'(x) < 0, \forall\, x \in (a, b)$ (B) $f'(x) > 0, \forall\, x \in (a, b)$ (C) $f'(x) = 0, \forall\, x \in (a, b)$ (D) $f(x) > 0, \forall\, x \in (a, b)$
35.	The derivative of $\sin(x^2)$ w.r.t. x, at $x = \sqrt{\pi}$ is : (A) 1 (B) -1 (C) $-2\sqrt{\pi}$ (D) $2\sqrt{\pi}$
36.	Check whether the function $f(x) = x^2\,\lvert x \rvert$ is differentiable at x = 0 or not.
37.	If $y = \sqrt{\tan\sqrt{x}}$, prove that $\sqrt{x}\,\dfrac{dy}{dx} = \dfrac{1+y^4}{4y}$.
38.	

39.	If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.
40.	If $y = (\tan x)^x$, then find $\frac{dy}{dx}$.
41.	$\frac{d}{dx}[\cos(\log x + e^x)]$ at $x = 1$ is : (A) $-\sin e$ (B) $\sin e$ (C) $-(1+e)\sin e$ (D) $(1+e)\sin e$
42.	The number of points of discontinuity of $f(x) = \begin{cases} \lvert x\rvert+3, & \text{if } x \le -3 \\ -2x, & \text{if } -3<x<3 \\ 6x+2, & \text{if } x \ge 3 \end{cases}$ is : (A) 0 (B) 1 (C) 2 (D) infinite
43.	Check the differentiability of $f(x) = \lvert\cos x\rvert$ at $x = \frac{\pi}{2}$.
44.	If $y = A\sin 2x + B\cos 2x$ and $\frac{d^2y}{dx^2} - ky = 0$, find the value of k.
45.	If $x^{30}y^{20} = (x+y)^{50}$, prove that $\frac{dy}{dx} = \frac{y}{x}$.
46.	Find $\frac{dy}{dx}$, if $5^x + 5^y = 5^{x+y}$.

47.	If $y = \sin^{-1} x$, then $\frac{d^2y}{dx^2}$ is : (A) $\sec y$ (B) $\sec y \tan y$ (C) $\sec^2 y \tan y$ (D) $\tan^2 y \sec y$
48.	A function $f(x) = \lvert 1 - x + \lvert x \rvert \rvert$ is : (A) discontinuous at $x = 1$ only (B) discontinuous at $x = 0$ only (C) discontinuous at $x = 0, 1$ (D) continuous everywhere
49.	A function $f(x) = \lvert 1 - x + \lvert x \rvert \rvert$ is : (A) discontinuous at $x = 1$ only (B) discontinuous at $x = 0$ only (C) discontinuous at $x = 0, 1$ (D) continuous everywhere
50.	If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$
51.	Check the differentiability of $f(x) = \begin{cases} x^2 + 1, & 0 \le x < 1 \\ 3 - x, & 1 \le x \le 2 \end{cases}$ at $x = 1$.
52.	If $x \cos(p + y) + \cos p \sin(p + y) = 0$, prove that $\cos p \frac{dy}{dx} = -\cos^2(p + y)$, where p is a constant.
53.	Find the value of a and b so that function f defined as : $f(x) = \begin{cases} \dfrac{x-2}{\lvert x-2 \rvert} + a, & \text{if } x < 2 \\ a + b, & \text{if } x = 2 \\ \dfrac{x-2}{\lvert x-2 \rvert} + b, & \text{if } x > 2 \end{cases}$ is a continuous function.

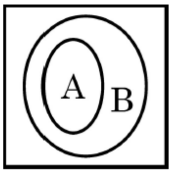
54.	Derivative of e^{2x} with respect to e^x, is : (A) e^x (B) $2e^x$ (C) $2e^{2x}$ (D) $2e^{3x}$
55.	For what value of k, the function given below is continuous at $x = 0$? $f(x) = \begin{cases} \dfrac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ (A) 0 (B) $\frac{1}{4}$ (C) 1 (D) 4
56.	If $y = \cos^3 (\sec^2 2t)$, find $\dfrac{dy}{dt}$.
57.	If $x^y = e^{x-y}$, prove that $\dfrac{dy}{dx} = \dfrac{\log x}{(1+\log x)^2}$.
58.	Given that $y = (\sin x)^x \cdot x^{\sin x} + a^x$, find $\dfrac{dy}{dx}$.
59.	If $xe^y = 1$, then the value of $\dfrac{dy}{dx}$ at $x = 1$ is : (A) -1 (B) 1 (C) $-e$ (D) $-\frac{1}{e}$
60.	Derivative of $e^{\sin^2 x}$ with respect to $\cos x$ is : (A) $\sin x \, e^{\sin^2 x}$ (B) $\cos x \, e^{\sin^2 x}$ (C) $-2 \cos x \, e^{\sin^2 x}$ (D) $-2 \sin^2 x \cos x \, e^{\sin^2 x}$

61.	Verify whether the function f defined by $$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & , \ x \neq 0 \\ 0 & , \ x = 0 \end{cases}$$ is continuous at $x = 0$ or not.
62.	Check for differentiability of the function f defined by $f(x) = \lvert x - 5 \rvert$, at the point $x = 5$.
63.	Find $\frac{dy}{dx}$, if $(\cos x)^y = (\cos y)^x$.
64.	If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.
65.	If $x = a \sin^3 \theta$, $y = b \cos^3 \theta$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.
66.	If $f(x) = \lvert x \rvert + \lvert x - 1 \rvert$, then which of the following is correct ? (A) $f(x)$ is both continuous and differentiable, at $x = 0$ and $x = 1$. (B) $f(x)$ is differentiable but not continuous, at $x = 0$ and $x = 1$. (C) $f(x)$ is continuous but not differentiable, at $x = 0$ and $x = 1$. (D) $f(x)$ is neither continuous nor differentiable, at $x = 0$ and $x = 1$.
67.	**Assertion (A)** : $f(x) = \begin{cases} 3x - 8, & x \leq 5 \\ 2k \ , & x > 5 \end{cases}$ is continuous at $x = 5$ for $k = \frac{5}{2}$. **Reason (R)** : For a function f to be continuous at $x = a$, $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$.
68.	Differentiate $2^{\cos^2 x}$ w.r.t $\cos^2 x$.
69.	If $\tan^{-1}(x^2 + y^2) = a^2$, then find $\frac{dy}{dx}$.

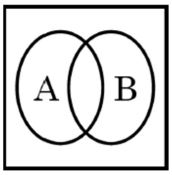
70. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then prove that $\dfrac{dy}{dx} = \sqrt{\dfrac{1-y^2}{1-x^2}}$.

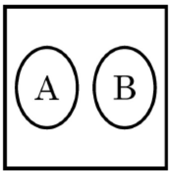
71. If $x = a\left(\cos\theta + \log\tan\dfrac{\theta}{2}\right)$ and $y = \sin\theta$, then find $\dfrac{d^2y}{dx^2}$ at $\theta = \dfrac{\pi}{4}$.

72. If A denotes the set of continuous functions and B denotes set of differentiable functions, then which of the following depicts the correct relation between set A and B ?

(A) 

(B) 

(C) 

(D) 

73. If $x = e^{\frac{x}{y}}$, then prove that $\dfrac{dy}{dx} = \dfrac{x-y}{x\log x}$.

74. If $f(x) = \begin{cases} 2x-3, & -3 \le x \le -2 \\ x+1, & -2 < x \le 0 \end{cases}$

Check the differentiability of $f(x)$ at $x = -2$.

75. If $y = \log\left(\sqrt{x} + \dfrac{1}{\sqrt{x}}\right)^2$, then show that $x(x+1)^2 y_2 + (x+1)^2 y_1 = 2$.

76. If $x\sqrt{1+y} + y\sqrt{1+x} = 0, -1 < x < 1, x \ne y$, then prove that $\dfrac{dy}{dx} = \dfrac{-1}{(1+x)^2}$.

77. If $f(x) = \begin{cases} 3x-2, & 0 < x \le 1 \\ 2x^2 + ax, & 1 < x < 2 \end{cases}$ is continuous for $x \in (0, 2)$, then a is equal to :

(A) -4 (B) $-\dfrac{7}{2}$

(C) -2 (D) -1

78. If $y = \log_{2x}(\sqrt{2x})$, then $\frac{dy}{dx}$ is equal to :

(A) 0 (B) 1

(C) $\frac{1}{x}$ (D) $\frac{1}{\sqrt{2x}}$

79. The function f defined by

$$f(x) = \begin{cases} x, & \text{if } x \le 1 \\ 5, & \text{if } x > 1 \end{cases}$$

is ***not*** continuous at :

(A) $x = 0$ (B) $x = 1$

(C) $x = 2$ (D) $x = 5$

80. Find k so that

$$f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x+1}, & x \neq -1 \\ k, & x = -1 \end{cases}$$

is continuous at $x = -1$.

81. Check the differentiability of function $f(x) = x|x|$ at $x = 0$.

82. If $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

is continuous at $x = 0$, then the value of a is :

(A) 1 (B) -1

(C) ± 1 (D) 0

83. If $f(x) = \{[x],\ x \in R\}$ is the greatest integer function, then the correct statement is :

(A) f is continuous but not differentiable at $x = 2$.

(B) f is neither continuous nor differentiable at $x = 2$.

(C) f is continuous as well as differentiable at $x = 2$.

(D) f is not continuous but differentiable at $x = 2$.

84. Differentiate $\dfrac{\sin x}{\sqrt{\cos x}}$ with respect to x.

85. If $y = 5 \cos x - 3 \sin x$, prove that $\dfrac{d^2y}{dx^2} + y = 0$.

86. For a positive constant 'a', differentiate $a^{t+\frac{1}{t}}$ with respect to $\left(t+\dfrac{1}{t}\right)^a$, where t is a non-zero real number.

87. Find $\dfrac{dy}{dx}$ if $y^x + x^y + x^x = a^b$, where a and b are constants.

88. If $f(x) = \begin{cases} \dfrac{\log(1+ax)+\log(1-bx)}{x}, & \text{for } x \neq 0 \\ k, & \text{for } x = 0 \end{cases}$

is continuous at $x = 0$, then the value of k is :

(A) a (B) $a + b$

(C) $a - b$ (D) b

89. If $y = a \cos(\log x) + b \sin(\log x)$, then $x^2y_2 + xy_1$ is :

(A) $\cot(\log x)$ (B) y

(C) $-y$ (D) $\tan(\log x)$

90.	If $\tan^{-1}(x^2 - y^2) = a$, where 'a' is a constant, then $\frac{dy}{dx}$ is : (A) $\frac{x}{y}$ (B) $-\frac{x}{y}$ (C) $\frac{a}{x}$ (D) $\frac{a}{y}$
91.	Differentiate $\sqrt{e^{\sqrt{2x}}}$ with respect to $e^{\sqrt{2x}}$ for $x > 0$.
92.	If $(x)^y = (y)^x$, then find $\frac{dy}{dx}$.
93.	Differentiate $y = \sin^{-1}(3x - 4x^3)$ w.r.t. x, if $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$.
94.	Differentiate $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ with respect to x, when $x \in (0, 1)$.
95.	If $f(x) = -2x^8$, then the correct statement is : (A) $f'\left(\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right)$ (B) $f'\left(\frac{1}{2}\right) = -f'\left(-\frac{1}{2}\right)$ (C) $-f'\left(\frac{1}{2}\right) = f\left(-\frac{1}{2}\right)$ (D) $f\left(\frac{1}{2}\right) = -f\left(-\frac{1}{2}\right)$
96.	If $f(x) = \begin{cases} 3ax - b & , x > 1 \\ 11 & , x = 1 \\ -5ax - 2b & , x < 1 \end{cases}$ is continuous at x = 1, then the values of a and b are : (A) $a = 3,\ b = 5$ (B) $a = 8,\ b = -1$ (C) $a = 1,\ b = -8$ (D) $a = -3,\ b = 5$

97.	*Assertion (A) :* $f(x) = \begin{cases} x \sin \frac{1}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ is continuous at x = 0. *Reason (R) :* When $x \to 0$, $\sin \frac{1}{x}$ is a finite value between – 1 and 1.
98.	Differentiate $\left(\frac{5^x}{x^5}\right)$ with respect to x.
99.	If $-2x^2 - 5xy + y^3 = 76$, then find $\frac{dy}{dx}$.
100	Differentiate $\log(x^x + \operatorname{cosec}^2 x)$ with respect to x.
	CASE STUDY
101	Let f(x) be a real valued function. Then its • Left Hand Derivative (L.H.D.) : $Lf'(a) = \lim\limits_{h \to 0} \frac{f(a-h)-f(a)}{-h}$ • Right Hand Derivative (R.H.D.) : $Rf'(a) = \lim\limits_{h \to 0} \frac{f(a+h)-f(a)}{h}$ Also, a function f(x) is said to be differentiable at x = a if its L.H.D. and R.H.D. at x = a exist and both are equal. For the function $f(x) = \begin{cases} \lvert x-3 \rvert, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$ answer the following questions : (i) What is R.H.D. of f(x) at x = 1 ? (ii) What is L.H.D. of f(x) at x = 1 ? (iii) (a) Check if the function f(x) is differentiable at x = 1. **OR** (iii) (b) Find $f'(2)$ and $f'(-1)$.
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